

Quantum circuit for implementing AES S-box with low costs

Hui-Nan Chen, Bin-Bin Cai, Fei Gao, Song Lin

Abstract—The Advanced Encryption Standard (AES) is widely used and well-studied for its efficiency and strong security. This paper presents quantum circuit designs for the AES S-box by introducing the composite field $F((2^4)^2)$ to replace the traditional field $F(2^8)$, enabling the inversion to be decomposed into operations over $F(2^4)$. This work reduces the quantum resource overhead required for implementing the S-box by decreasing the number of *CNOT* gates in the matrix multiplication, lowering the depth of *T* gates in both the inversion circuit and the multiplication circuit. Besides, the widths for the S-box quantum circuits are also optimized during the inversion circuit and multiplication circuit. With a linear key schedule, the resulting AES-128 quantum circuit reduce the product of the circuit width and *T* depth to 102800, which is the lowest known to date.

Index Terms—AES, S-box, quantum circuit, resource estimation, composite field arithmetic.

I. INTRODUCTION

IN recent years, the deep integration of quantum mechanics and information science has driven the rapid advancement of quantum information [1], [2], positioning it as a frontier field in scientific research and technological innovation. As a vital branch of this field, quantum computing has attracted widespread attention due to its unique computational model and powerful parallel processing capabilities. Unlike traditional computers that rely on binary bit states, quantum computers leverage the principles of superposition and entanglement, allowing qubits to process multiple states simultaneously and achieve unprecedented parallel computing capabilities [3], [4], [5], [6]. This grants quantum computing the potential to surpass classical computation, especially by offering significant speedup in solving certain complex problems [7], [8], [9]. In recent years, ongoing progress in areas, such as quantum architecture search [10], variational

quantum algorithm [11], and quantum neural network [12], has significantly broadened the scope of quantum computing in learning systems and secure information processing. This paradigm shift has profoundly impacted various disciplines, among which cryptography is one of the most significantly affected.

In the evaluation of cryptographic system security, quantum algorithms offer unprecedented acceleration in the attack process. Representative examples include Shors algorithm [13], [14], Grovers algorithm [15], [16], and Simons algorithm [17], [18]. Shor's algorithm leverages quantum computation to determine the periodicity of a function and employs the quantum Fourier transform [19], [20], [21] to efficiently solve the factoring problem, significantly accelerating the decomposition of large integers. Compared with the exponential complexity of classical algorithms, Shor's algorithm completes this task in polynomial time. As a result, it poses a direct threat to classical encryption systems based on the hardness of integer factorization, particularly the widely used RSA [22]. Grovers algorithm does not pose a direct threat to public-key cryptosystems as Shors algorithm, while it has significant implications for the security of symmetric encryption algorithms, such as AES [23]. In the classical computing environment, performing an exhaustive key search on a symmetric cipher with an n -bit key typically requires $O(2^n)$ operations. However, when Grovers algorithm is applied in a quantum computing context, the complexity is reduced to $O(2^{n/2})$, achieving a quadratic speedup. Simons algorithm is one of the earliest quantum algorithms to demonstrate exponential speedup over classical counterparts. In recent years, it has shown strong capability in identifying hidden periodic structures within symmetric cryptographic constructions, such as the EvenCMansour cipher [24] and Feistel networks [25]. Although these theoretical advancements demonstrate the immense potential of quantum computing in cryptanalysis, a critical challenge remains in practical attacks, that is, how to design and implement the quantum *Oracle* which is tailored to specific encryption algorithms. Therefore, constructing an efficient and feasible quantum *Oracle* has become a key challenge in advancing quantum cryptanalysis from theory to practice.

As a core component of quantum algorithms, the *Oracle* serves to encapsulate the target function into a quantum subcircuit that can be queried by the algorithm, enabling effective access to and manipulation of the internal structure of cryptographic schemes. However, designing an *Oracle* is often highly complex, as it requires faithfully replicating the classical functions logic while also carefully managing key quantum resources, such as the depth of the quantum circuit and the number of qubits [26], [27]. Therefore, minimizing

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the resource overhead of quantum circuit implementations while maintaining functional correctness has become one of the central challenges in quantum cryptanalysis [28]. Resource estimation and optimization around *Oracle* construction have emerged as critical steps toward the practical realization of quantum attack strategies.

As one of the most widely used symmetric encryption standards, AES has naturally become a core target in quantum cryptanalysis. Quantum resources required by Grover's algorithm to attack AES have been used as a reference benchmark for evaluating the security strength of post-quantum cryptographic algorithms in 2016 post-quantum standardization call of the National Institute of Standards and Technology (NIST) [29], [30]. Therefore, constructing optimized quantum circuits for AES is of critical importance for accurately assessing the resource requirements of Grover search attack. To achieve such optimization, it is necessary to implement a quantum *Oracle* for AES and perform quantum resource estimation and refinement. Previous works on AES quantum circuit design have primarily focused on three directions: reducing the width (i.e., the number of qubits), the depth, or the product of width and depth.

To reduce the quantum resource requirements of Grover search attack, researchers have focused primarily on reducing the width of the quantum circuit of AES. In 2016, Grassl et al. [31] designed a quantum circuit for the S-box with a width of 40, using a zig-zag approach. They constructed a quantum circuit for AES-128 with a final circuit width of 984. In 2018, Almazrooe et al. [32] optimized the key expansion strategy, further reducing the width of AES-128 to 976. In 2020, Langenberg et al. [27] proposed a quantum circuit for the S-box with a width of 32, incorporating a zig-zag structure in the key expansion, which led to a reduction in the width of AES-128 to 864. In the same year, Zou et al. [33] designed a quantum circuit for the S-box with a width of 22 and combined it with an improved zig-zag method with a width of 23, leading to the implementation of quantum circuit for AES-128 with a width of 512. In 2022, Wang et al. [34] presented a straight-line iterative method for key expansion, reducing the width of AES-128 to 400. Later, Li et al. [35] further minimized the width of AES-128, reducing it to 270 by constructing the quantum circuit for the S-box with a width of 22 and applying a streamlining optimization strategy. In 2023, Li et al. [36] further improved the previous quantum circuits, reducing the width of AES-128 to 264. In 2025, Huang et al. [37] proposed a quantum circuit for AES-128 that requires only 256 qubits and does not rely on any additional ancillary qubits.

In addition to width, depth of the quantum circuit for AES also plays a crucial role in determining the feasibility of quantum attacks. Therefore, considerable effort has been devoted to optimizing the sequence of gate operations to a lower depth. Jang et al. [38] proposed the first S-box quantum circuit designed to reduce the depth. In their implementation, the AES-128 quantum circuit achieved a T -depth of 120. Huang and Sun [39], based on the classical circuit for the S-box proposed by Boyar and Peralta [40], constructed a quantum circuit of the S-box focused on optimizing the depth of T gates. Their design reduced the T -depth of the AES-128

quantum circuit to 60.

Furthermore, the cost of quantum attacks is also evaluated by the product of width and depth [26]. The metric was first proposed by Jaques et al. [26] and refined by Jang et al. [38]. In 2023, Liu et al. [41] proposed a technique called m-XOR to identify reusable qubits. They also designed a compact quantum circuit for S-box and ultimately constructed an AES-128 quantum circuit with the $DW(T)$ value of 147560. In 2025, Jiang et al. [42] applied the controlled control qubit cascade (CCQC) technique to quantum circuit construction and built a quantum circuit for AES-128 with the $DW(T)$ value of 128640.

Contribution In this paper, we refine the method proposed in Ref.[35] and develop a quantum implementation of AES with lower resource overhead, reducing the the product of the quantum circuit width and T depth (i.e., $DW(T)$). The approach analyzes the algebraic structure of the S-box and introduces the composite field $F((2^4)^2)$, which is isomorphic to the finite field $F(2^8)$. This transformation allows the inversion of elements in $F(2^8)$ to be performed within $F((2^4)^2)$. Furthermore, the inversion in $F((2^4)^2)$ is broken down into a sequence of operations over the smaller finite field $F(2^4)$, which facilitates a quantum circuit design for the S-box with the reduced $DW(T)$ value. In the process mentioned above, we optimize the quantum implementation of matrix multiplication in $F(2^4)$ by reducing the number of $CNOT$ gates, lowering the number of T gates and T depth in the inversion circuit in $F(2^4)$, and achieving the quantum multiplication circuit in $F(2^4)$ with the reduced width and depth. Therefore, we obtain the currently lowest known $DW(T)$ value for AES-128, which is 102800. Our design shows significant advantages under the MAXDEPTH parameter, proposed by NIST in 2016 [29], [30] to evaluate quantum circuit depth and attack cost. Compared with the work in Ref.[42], the $DW(T)$ value in our quantum circuit of AES-128 is reduced by 20.09%.

II. PRELIMINARIES

In this section, we provide an overview of the AES block cipher, composite field arithmetic, and some basic quantum gates.

A. The AES block cipher

AES uses a 128-bit block size and supports key lengths of 128, 192, and 256 bits, known as AES-128, AES-192, and AES-256, respectively. The number of encryption rounds N_R for AES is 10, 12, and 14 for each key length. The plaintext M undergoes one AddRoundKey operation, followed by N_R rounds of the round function, ultimately producing the ciphertext C . Each round function consists of SubBytes, ShiftRows, MixColumns and AddRoundKey, except for the final round function, which contains only SubBytes, ShiftRows, and AddRoundKey.

1) *The S-box*: SubBytes, which is denoted as $S(a)$, represents the only nonlinear transformation within AES. It is performed by applying the same S-box transformation to each byte individually. The S-box is an 8×8 lookup table that maps an input byte to an output byte. $S(a)$ takes a byte $a \in F(2^8)$ as

input, where $a = (a_7 a_6 a_5 a_4 a_3 a_2 a_1 a_0)$ can be represented by the polynomial $(a_7x^7 + a_6x^6 + a_5x^5 + a_4x^4 + a_3x^3 + a_2x^2 + a_1x + a_0)$ with coefficients $\{0, 1\}$.

To implement the quantum circuit implementation of the S-box, we primarily focus on its algebraic representation. The process begins by computing the multiplicative inverse of a , denoted as a^{-1} , followed by an affine transformation that involves matrix multiplication and modular addition with a constant vector. Consequently, the algebraic form of $S(a)$ can be expressed as

$$S(a) = Aa^{-1} \oplus c, \quad (1)$$

where the matrix A and vector c are defined as

$$A = \begin{pmatrix} 1 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 1 & 1 & 0 & 0 & 0 & 1 & 1 & 1 \\ 1 & 1 & 1 & 0 & 0 & 0 & 1 & 1 \\ 1 & 1 & 1 & 1 & 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 & 1 \end{pmatrix}, \quad c = \begin{pmatrix} 1 \\ 1 \\ 0 \\ 0 \\ 0 \\ 1 \\ 1 \\ 0 \end{pmatrix}.$$

2) *The key expansion process:* AES-128, AES-192, and AES-256 have different key expansion processes due to their varying initial key lengths. Considering 32 bits as a word, the initial key K_0 can be divided into N_K words for AES-128, AES-192 and AES-256, where N_K equals 4, 6 and 8. For AES-128, the initial key can be represented as $K_0 = W_3W_2W_1W_0$. After N_R round key expansion process, $(N_R + 1)$ subkeys are generated. The subkey for the i -th round of AES-128, AES-192, and AES-256 are all denoted as $K_i = W_{4i+3}W_{4i+2}W_{4i+1}W_{4i}$, where $0 \leq i \leq N_R$. Each subkey is 128 bits in length. The key expansion process for AES-128 and AES-192 is shown in Algorithm 1, while the key expansion process for AES-256 is shown in Algorithm 2. In these algorithms, *RotWord* represents a circular left shift of a word by 8 bits, and *Rcon* $[j]$ denotes that *XOR* the current state with a round constant, where $j \geq 0$. Each round constant *Rcon* $[j]$ is defined as a four-byte word of the form $(RC[j], 0x00, 0x00, 0x00)$, where only the first byte *RC* $[j]$ is nonzero and the remaining three bytes are zero. The value of *RC* $[j]$ corresponds to the element x^{j-1} in the finite field $F(2^8)$, reduced modulo the irreducible polynomial $m(x) = x^8 + x^4 + x^3 + x + 1$, which serves as the generator polynomial in AES.

Algorithm 1 The key expansion process for AES-128 and AES-192

```

for  $t$  from  $N_K$  to  $4 \times (N_R + 1)$  do
  if  $t \bmod 4 == 0$  then
     $W_t = W_{t-4} \oplus \text{SubBytes}(\text{RotWord}(W_{t-1})) \oplus$ 
       $\text{Rcon}[t/4]$ 
  else
     $W_t = W_{t-4} \oplus W_{t-1}$ 
  end if
end for

```

Algorithm 2 The key expansion process for AES-256

```

for  $t$  from  $N_K$  to  $4 \times (N_R + 1)$  do
  if  $t \bmod 8 == 0$  then
     $W_t = W_{t-8} \oplus \text{SubBytes}(\text{RotWord}(W_{t-1})) \oplus$ 
       $\text{Rcon}[t/8]$ 
  else if  $t \bmod 8 == 4$  then
     $W_t = W_{t-8} \oplus \text{SubBytes}(\text{RotWord}(W_{t-1}))$ 
  else
     $W_t = W_{t-8} \oplus W_{t-1}$ 
  end if
end for

```

B. Composite field arithmetic

In combinatorial domain operations, elements in higher-order domains can be expressed as linear combinations of elements in lower-order domains, where operations in lower-order domains are simpler and less costly. Therefore, solving the multiplicative inverse of any element a in a finite field can be transformed into solving the multiplicative inverse of the corresponding element in a combinatorial domain using a mapping matrix M . After obtaining the multiplicative inverse of the corresponding element in the combinatorial domain, the multiplicative inverse of a can be derived using the inverse matrix M^{-1} , resulting in a^{-1} .

Wolkerstorfer et al. [43] proposed an implementation of the S-box using the composite fields $F((2^4)^2)$ and $F(2^4)$, i.e.,

$$\begin{cases} F((2^4)^2) : x^2 + x + \lambda, \\ F(2^4) : x^4 + x + 1, \end{cases}$$

where

$$\lambda = x^3 + x^2 + x \in F(2^4). \quad (2)$$

The mapping matrix $M : F(2^8) \rightarrow F((2^4)^2)$ and the inverse matrix $M^{-1} : F((2^4)^2) \rightarrow F(2^8)$ are defined as

$$M = \begin{pmatrix} 1 & 0 & 0 & 0 & 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 \end{pmatrix},$$

$$M^{-1} = \begin{pmatrix} 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 0 & 1 \end{pmatrix}.$$

It follows that the algebraic expression of Eq.(1) can be rewritten as

$$S(x) = AM^{-1}(Ma)^{-1} \oplus c, \quad (3)$$

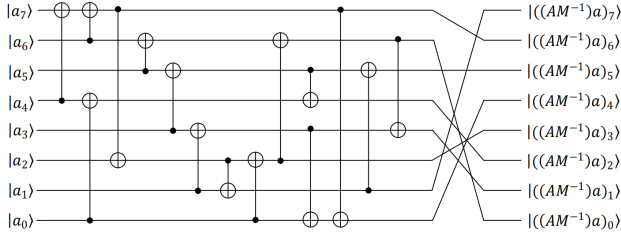
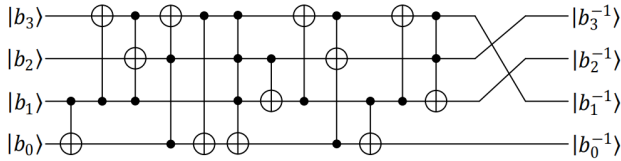
Fig. 4. The quantum circuit for $U_{AM^{-1}} : |a\rangle \rightarrow |(AM^{-1})a\rangle$.

TABLE I
THE INPUT ELEMENTS AND CORRESPONDING INVERSE OUTPUT
ELEMENTS IN $F(2^4)$

$\mathbf{b}$	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
$\mathbf{b}^{-1}$	0	1	9	E	D	B	7	6	F	2	C	5	A	4	3	8

in $F(2^4)$, with its corresponding multiplicative inverse $\mathbf{b}^{-1}$ shown in Table I.

In this paper, we use the automated tool DORCIS [45] to realize the quantum circuit of the 4×4 lookup table, which solves the multiplicative inverse of elements in $F(2^4)$. The resulting quantum circuit is denoted as $F(2^4)_{inv} : |\mathbf{b}\rangle \rightarrow |\mathbf{b}^{-1}\rangle$, where $\mathbf{b} = (b_3 \ b_2 \ b_1 \ b_0)$ is the input element and $\mathbf{b}^{-1} = (b_3^{-1} \ b_2^{-1} \ b_1^{-1} \ b_0^{-1})$ is the output element, as shown in Figure 5.

Fig. 5. The quantum circuit for $F(2^4)_{inv} : |\mathbf{b}\rangle \rightarrow |\mathbf{b}^{-1}\rangle$.

The 3-controlled NOT gate appearing in $F(2^4)_{inv}$ (shown in Figure 5) can be implemented using the decomposition method $|a\rangle|b\rangle|c\rangle|d\rangle|0\rangle \rightarrow |a\rangle|b\rangle|c\rangle|d \oplus abc\rangle|0\rangle$ (shown in Figure 6), which requires a clean auxiliary qubit, resulting in the decomposed circuit $F(2^4)_{inv0} : |\mathbf{b}\rangle|0\rangle \rightarrow |\mathbf{b}^{-1}\rangle|0\rangle$.

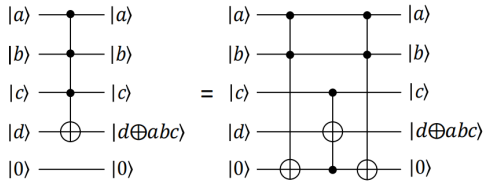


Fig. 6. The decomposition of 3-controlled NOT gate with a clean auxiliary qubit.

Furthermore, to further reduce the depth of *Toffoli* gates, we derive the algebraic expression of $F(2^4)_{inv}$ (shown in Figure 5), i.e.,

TABLE II
QUANTUM RESOURCE ESTIMATION FOR MULTIPLICATIVE INVERSION IN $F(2^4)$. #qubits DENOTES QUBIT COUNT; #Toffoli, #CNOT, AND #NOT ARE THE COUNTS OF RESPECTIVE GATES.

	#qubits	#Toffoli	#CNOT	#NOT	Toffoli depth
$F(2^4)_{inv0}$	5	7	7	0	7
$F(2^4)_{inv1}$	25	20	42	0	4

TABLE III
QUANTUM RESOURCE ESTIMATION FOR MULTIPLICATIVE INVERSION IN $F(2^4)$ AFTER DECOMPOSING THE *Toffoli* GATES INTO COMBINATIONS OF *T* GATES AND *Clifford* GATES. #T AND #Clifford ARE THE COUNTS OF RESPECTIVE GATES.

	#qubits	#T	#Clifford	T depth
$F(2^4)_{inv0}$	7	24	129	6
$F(2^4)_{inv1}$	29	40	222	2

$$\begin{aligned}
 b_3^{-1} &= b_3 + b_2 + b_1 + b_3b_2 + b_3b_1 + b_3b_0 + b_3b_2b_1, \\
 b_2^{-1} &= b_3 + b_2 + b_3b_0 + b_2b_0 + b_1b_0 + b_3b_2b_0, \\
 b_1^{-1} &= b_3 + b_3b_1 + b_2b_1 + b_2b_0 + b_1b_0 + b_3b_1b_0, \\
 b_0^{-1} &= b_3 + b_2 + b_1 + b_0 + b_2b_1 + b_2b_0 + b_3b_2b_1 + b_2b_1b_0.
 \end{aligned} \tag{6}$$

Based on Eq.(6), the quantum circuit can be redesigned in Figure 7 and is denoted as $F(2^4)_{inv1} : |\mathbf{b}\rangle|0\rangle|0\rangle \rightarrow |\mathbf{b}^{-1}\rangle|\mathbf{b}\rangle|0\rangle$.

The quantum resource estimation for implementing the multiplicative inverse in $F(2^4)$ is summarized in Table II, and the quantum resource estimation after decomposing the *Toffoli* gates into combinations of *T* gates and *Clifford* gates is summarized in Table III.

C. Quantum circuit implementation of $q^2 \times \lambda$

The element q in $F(2^4)$ can be written as $q = q_3x^3 + q_2x^2 + q_1x + q_0$, where q_i ($i \in 0, 1, 2, 3$) is an element in $F(2^4)$ and $\lambda = x^3 + x^2 + x \in F(2^4)$ is defined in Eq.(2). Through a calculation, $q^2 \times \lambda$ is expressed as

$$q^2 \times \lambda = (q_1 + q_0)x^3 + (q_3 + q_1 + q_0)x^2 + q_0x + (q_2 + q_1). \tag{7}$$

Based on Eq.(7), we can derive a quantum circuit for $U_{q^2 \times \lambda} : |q\rangle \rightarrow |q^2 \times \lambda\rangle$ in Figure 8. The $U_{q^2 \times \lambda}$ can be implemented with 4 qubits and 3 *CNOT* gates.

D. Quantum circuit implementations of multiplication in $F(2^4)$

In $F(2^4)$, any element of a and b can be written as

$$\begin{cases} a = a_3y^3 + a_2y^2 + a_1y + a_0, \\ b = b_3y^3 + b_2y^2 + b_1y + b_0. \end{cases}$$

$a \cdot b$ is expressed as

$$a \cdot b = (ab)_3y^3 + (ab)_2y^2 + (ab)_1y + (ab)_0, \tag{8}$$

where $(ab)_i$ ($i \in 0, 1, 2, 3$) is the i -th term of $a \cdot b$ and $(ab)_3 = (a_3 + a_2 + a_1 + a_0)(b_3 + b_2 + b_1 + b_0) + (a_3 + a_2)(b_3 +$

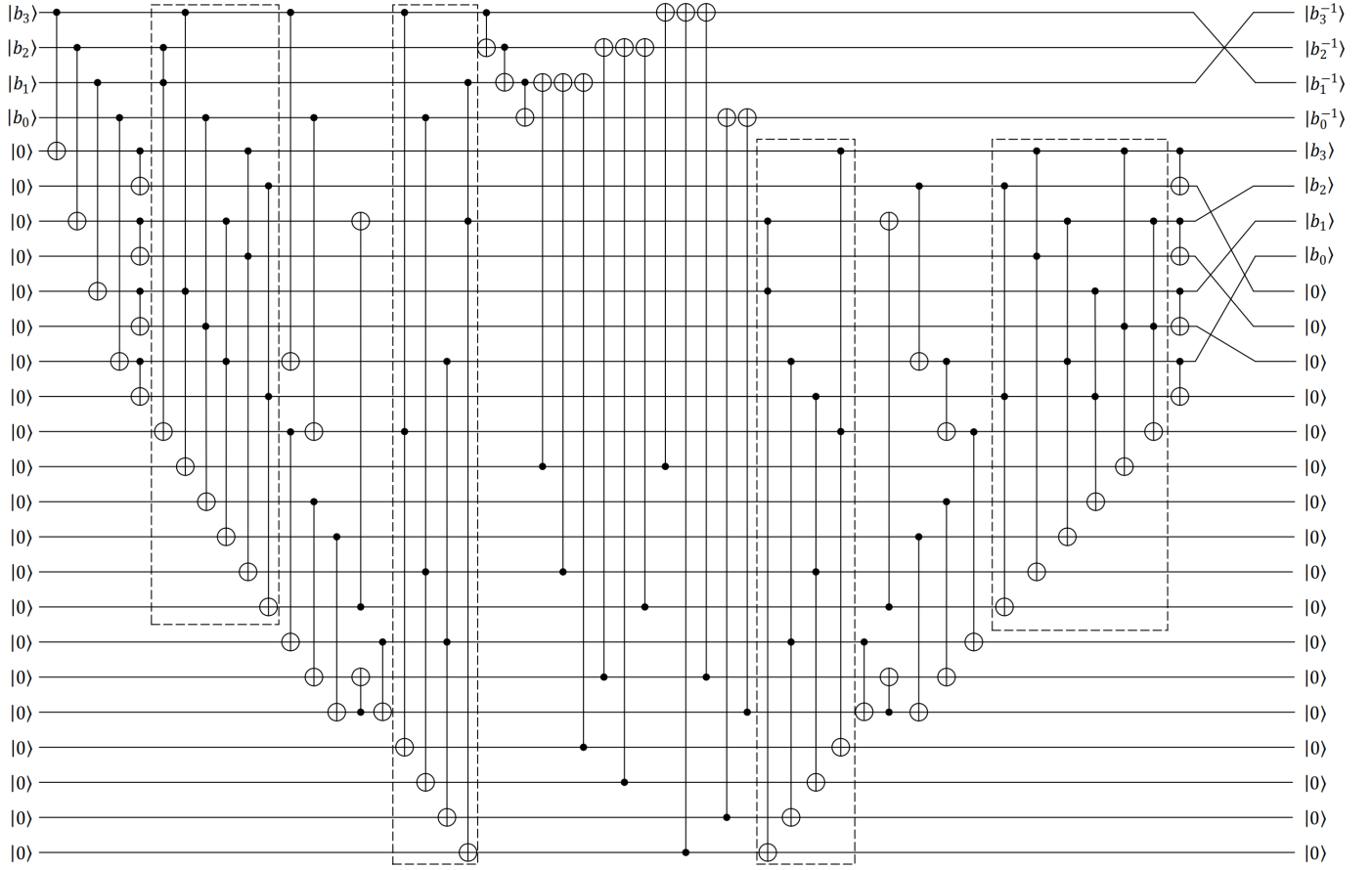


Fig. 7. The quantum circuit for $F(2^4)_{inv1} : |b\rangle|0\rangle|0\rangle \rightarrow |b^{-1}\rangle|b\rangle|0\rangle$, where dashed boxes indicate the *Toffoli* gates can be applied in parallel.

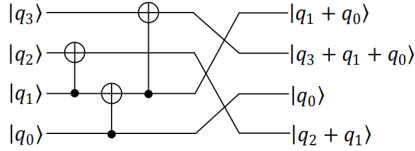


Fig. 8. The quantum circuit for $U_{q^2 \times \lambda} : |q\rangle \rightarrow |q^2 \times \lambda\rangle$.

$$\begin{aligned}
 & b_2) + (a_3 + a_1)(b_3 + b_1) + (a_2 + a_0)(b_2 + b_0) + (a_1 + a_0)(b_1 + b_0) + a_2b_2 + a_1b_1 + a_0b_0, \\
 (ab)_2 &= (a_3 + a_2)(b_3 + b_2) + (a_2 + a_0)(b_2 + b_0) + a_1b_1 + a_0b_0, \\
 (ab)_1 &= (a_3 + a_2)(b_3 + b_2) + (a_3 + a_1)(b_3 + b_1) + (a_1 + a_0)(b_1 + b_0) + a_0b_0, \\
 (ab)_0 &= (a_3 + a_1)(b_3 + b_1) + a_3b_3 + a_2b_2 + a_1b_1 + a_0b_0.
 \end{aligned}$$

Based on Eq.(8), we present a quantum circuit for $Mul_0 : |a\rangle|b\rangle|0\rangle|0\rangle \rightarrow |a\rangle|b\rangle|a \cdot b\rangle|0\rangle$ of multiplication in $F(2^4)$ in Figure 9. Mul_0 can be implemented with 28 qubits, 15 *Toffoli* gates and 55 *CNOT* gates. The *Toffoli* depth of Mul_0 is 2. After decomposing the *Toffoli* gates in Mul_0 , it is found that implementing Mul_0 requires 36 qubits, 36 *T* gates and 196 *Clifford* gates. The *T* depth of Mul_0 is 1. It is worth noting that $Mul_0^\dagger$ is the inverse operation of Mul_0 . After decomposing the *Toffoli* gates in $Mul_0^\dagger$ into *T* and *Clifford* gates, the implementation of $Mul_0^\dagger$ requires 36 qubits, 24 *T* gates, and 184 *Clifford* gates. The *T* depth of $Mul_0^\dagger$ is 1.

Since the key expansion process involves the *SubBytes* operation, the result of applying *SubBytes* to the current state must be XORed with another state while keeping the current state unchanged. In this case, Mul_0 is not applicable, so a new multiplication operation, $Mul_1 : |a\rangle|b\rangle|h\rangle|0\rangle \rightarrow |a\rangle|b\rangle|h + a \cdot b\rangle|0\rangle$, needs to be constructed. Mul_1 is shown in Figure 10, where Mul_0 denotes the quantum circuit shown in Figure 9. Mul_1 can be implemented with 28 qubits, 15 *Toffoli* gates and 59 *CNOT* gates. The *Toffoli* depth of Mul_1 is 2. After decomposing the *Toffoli* gates in Mul_1 , it is found that implementing Mul_1 requires 37 qubits, 36 *T* gates and 200 *Clifford* gates. The *T* depth of Mul_1 is 1.

During the encryption process of AES, the optimal implementation of *SubBytes* is to act directly on the qubits associated with the current state, which necessitates the design of an in-place quantum circuit for the S-box. Accordingly, the quantum circuit for multiplication in $F(2^4)$ should also be in-place. However, the previously designed multiplication circuits Mul_0 and Mul_1 perform the result on ancillary qubits, following an out-of-place design strategy. Therefore, referring to Ref.[35], we present an in-place quantum circuit for $Mul_2 : |a\rangle|b\rangle|0\rangle \rightarrow |a \cdot b\rangle|b\rangle|0\rangle$, as shown in Figure 11, where $F(2^4)_{inv0}$ denotes the quantum circuits in Section III-B, and $Mul_0^\dagger$ is the inverse process of Mul_0 .

Moreover, if five additional clean ancillary qubits can be introduced into Mul_2 (shown in Figure 11), $F(2^4)_{inv0}$ in

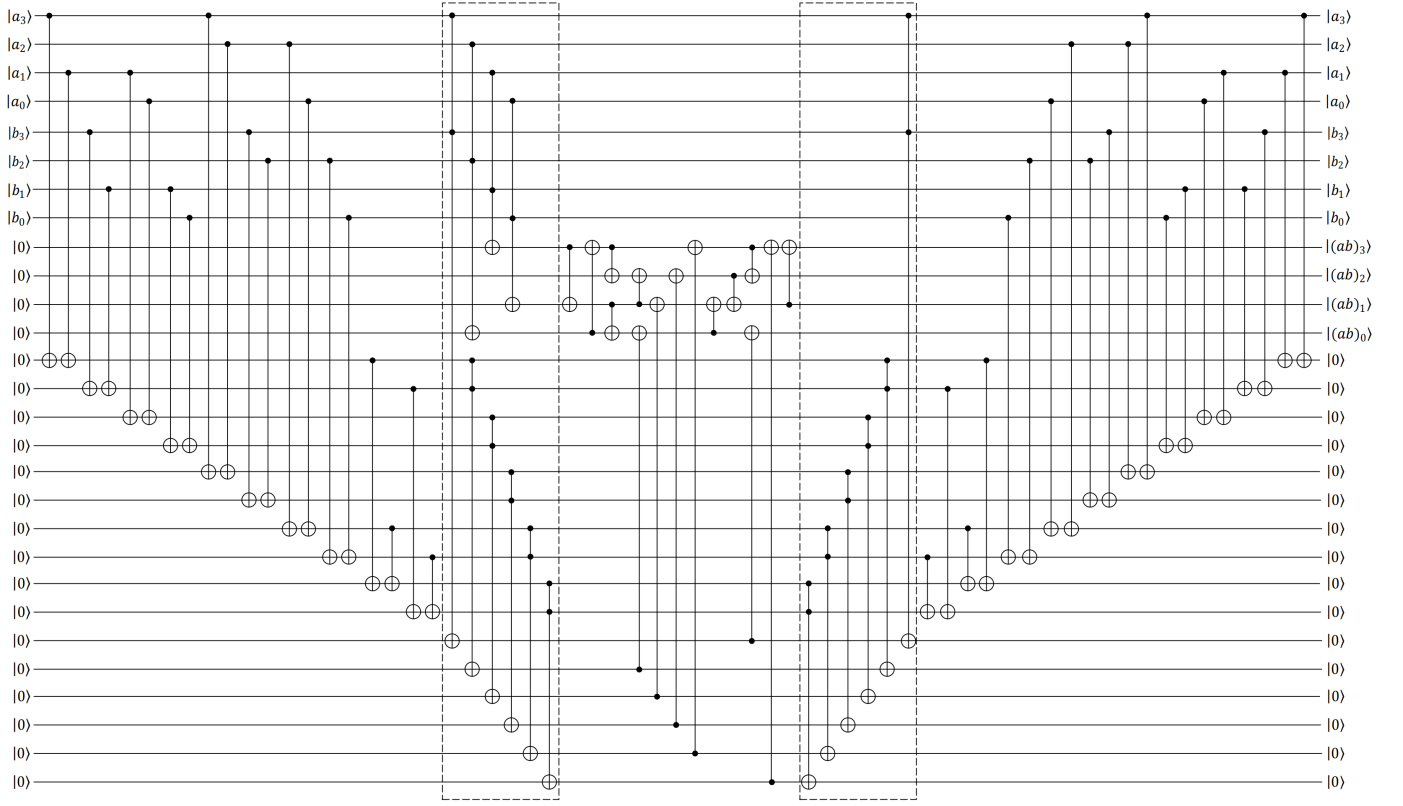


Fig. 9. The quantum circuit for $Mul_0 : |a\rangle|b\rangle|0\rangle|0\rangle \rightarrow |a\rangle|b\rangle|a \cdot b\rangle|0\rangle$.

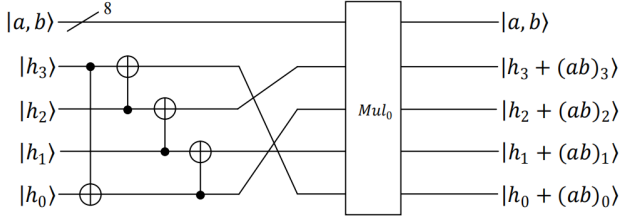


Fig. 10. The quantum circuit for $Mul_1 : |a\rangle|b\rangle|h\rangle|0\rangle \rightarrow |a\rangle|b\rangle|h + a \cdot b\rangle|0\rangle$.

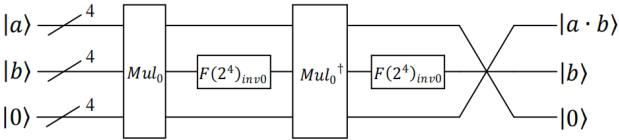


Fig. 11. The quantum circuit for $Mul_2 : |a\rangle|b\rangle|0\rangle \rightarrow |a \cdot b\rangle|b\rangle|0\rangle$.

Mul_2 can be replaced by $F(2^4)_{inv1}$ (shown in Figure 12), which increases the number of *Toffoli* gates by 26 but decreases the *Toffoli* depth by 8.

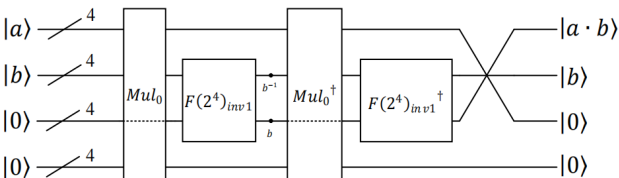


Fig. 12. The new quantum circuit for $Mul_2 : |a\rangle|b\rangle|0\rangle \rightarrow |a \cdot b\rangle|b\rangle|0\rangle$.

TABLE IV
QUANTUM RESOURCE ESTIMATION FOR IMPLEMENTING Mul_2 .

Operation	#qubits	#Toffoli	#CNOT	#NOT	Toffoli depth
Mul_2	28	44	124	0	18
	33	70	194	0	12

TABLE V
QUANTUM RESOURCE ESTIMATION FOR IMPLEMENTING Mul_2 AFTER DECOMPOSING THE *Toffoli* GATES INTO COMBINATIONS OF *T* GATES AND *Clifford* GATES.

Operation	#qubits	#T	#Clifford	T depth
Mul_2	36	108	638	14
	40	140	824	6

The quantum resources required for implementing Mul_2 are summarized in Table IV, and the quantum resource estimation after decomposing the *Toffoli* gates into combinations of *T* gates and *Clifford* gates is summarized in Table V.

E. The quantum circuits of S-box

This section presents three distinct quantum circuits for implementing the S-box, depending on varying target qubit states.

1) The quantum circuit for $c_1 : |a\rangle|0\rangle|0\rangle \rightarrow |a\rangle|S(a)\rangle|0\rangle$: When the target qubit is $|0\rangle$, the quantum circuit for $c_1 : |a\rangle|0\rangle|0\rangle \rightarrow |a\rangle|S(a)\rangle|0\rangle$ can be provided. Based on Eq.(3), the S-box can be computed by realizing the quantum circuits

of M , $(Ma)^{-1}$, AM^{-1} and adding a vector c in order. The quantum circuits of M and AM^{-1} have been presented in Section III-A. The addition of the vector c can be implemented with only 4 *NOT* gates. The remaining component is the quantum circuit that computes the multiplicative inverse $(Ma)^{-1}$ in $F((2^4)^2)$. Therefore, we first construct the quantum circuit for $U_{inv0} : |p\rangle|0\rangle|0\rangle \rightarrow |p\rangle|p^{-1}\rangle|0\rangle$ that implements the multiplicative inverse $(Ma)^{-1}$ in $F((2^4)^2)$, and the quantum circuit for c_1 is introduced subsequently.

Any element in $F((2^4)^2)$ can be denoted as $p = p_0 + p_1x$, where $p_0, p_1 \in F(2^4)$. The multiplicative inverse of p in $F((2^4)^2)$, denoted as p^{-1} , is represented as $p^{-1} = n_0 + n_1x$ for distinction. The value of p^{-1} can be determined through a four-part process involving $(p^{17})^{-1}(p_0 + p_1)$, $(p^{17})^{-1}p_1x$, $p_1^2 \times \lambda$, and $(p_0 + p_1)p_0$. Therefore, by combining the quantum circuits of $F(2^4)_{inv0}$ (see Section III-B), Mul_0 (see Figure 9), Mul_1 (see Figure 10), and $U_{q^2 \times \lambda}$ (see Figure 8), the quantum circuit $U_{inv0}^1 : |p\rangle|0\rangle|0\rangle \rightarrow |p\rangle|p^{-1}\rangle|0\rangle$ for computing the multiplicative inversion in $F((2^4)^2)$ can be constructed as shown in Figure 13. The operation $U_{(p^{-1})^{17}}^\dagger$ in Figure 13 is the inverse of $U_{(p^{-1})^{17}}$, where $U_{(p^{-1})^{17}}$ computes $(p^{-1})^{17}$ using n_0 and n_1 .

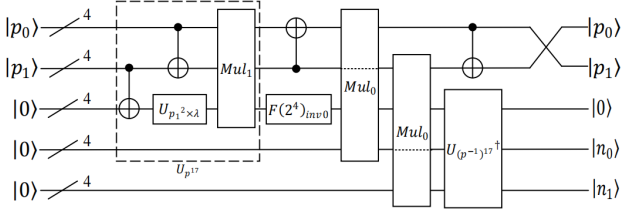


Fig. 13. The quantum circuit for $U_{inv0}^1 : |p\rangle|0\rangle|0\rangle \rightarrow |p\rangle|p^{-1}\rangle|0\rangle$, where the group of operations within the dashed box is collectively denoted as $U_{p^{17}}$, which is used to compute p^{17} from p_0 and p_1 .

If five clean ancillary qubits are added to the U_{inv0}^1 (as shown in Figure 13), the module $F(2^4)_{inv0}$ can be replaced with $F(2^4)_{inv1}$, resulting in U_{inv0}^2 (as shown in Figure 14).

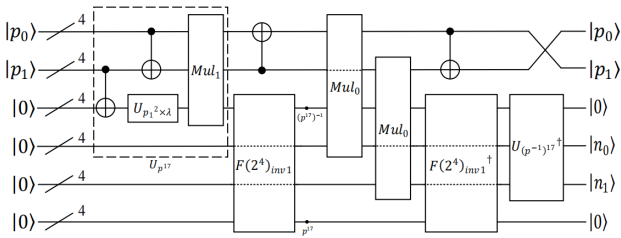


Fig. 14. The quantum circuit for $U_{inv0}^2 : |p\rangle|0\rangle|0\rangle \rightarrow |p\rangle|p^{-1}\rangle|0\rangle$.

By combining the quantum circuit of U_M (see Figure 3), U_{inv0} (U_{inv0} consists of U_{inv0}^1 shown in Figure 13 and U_{inv0}^2 shown in Figure 14), and $U_{AM^{-1}}$ (see Figure 4), the quantum circuit for $c_1 : |a\rangle|0\rangle|0\rangle \rightarrow |a\rangle|S(a)\rangle|0\rangle$ can be constructed as shown in Figure 15, where $U_M^\dagger$ is the inverse process of U_M .

The quantum resources required for implementing c_1 are summarized in Table VI, where c_1^1 indicates that the U_{inv0} used in c_1 is U_{inv0}^1 , while c_1^2 indicates that the U_{inv0} used in c_1 is U_{inv0}^2 . The quantum resource estimation after decomposing

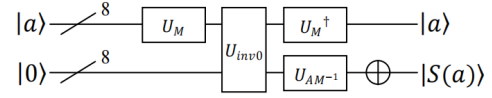


Fig. 15. The quantum circuit for $c_1 : |a\rangle|0\rangle|0\rangle \rightarrow |a\rangle|S(a)\rangle|0\rangle$.

TABLE VI
QUANTUM RESOURCE ESTIMATION FOR IMPLEMENTING c_1 .

Operation	#qubits	#Tofoli	#CNOT	#NOT	Tofoli depth
c_1^1	36	67	300	4	15
c_1^2	41	100	377	4	16

TABLE VII
QUANTUM RESOURCE ESTIMATION FOR IMPLEMENTING c_1 AFTER DECOMPOSING THE *Tofoli* GATES INTO COMBINATIONS OF *T* GATES AND *Clifford* GATES.

Operation	#qubits	#T	#Clifford	T depth
c_1^1	44	168	990	10
c_1^2	48	224	1305	8

the *Tofoli* gates into combinations of *T* gates and *Clifford* gates is summarized in Table VII.

2) The quantum circuit for $c_2 : |a\rangle|b\rangle|0\rangle \rightarrow |a\rangle|b \oplus S(a)\rangle|0\rangle$: When the target qubit is $|b\rangle$ instead of $|0\rangle$, the given quantum circuit can be denoted as $c_2 : |a\rangle|b\rangle|0\rangle \rightarrow |a\rangle|b \oplus S(a)\rangle|0\rangle$. By this time, the quantum circuit U_{inv0} (U_{inv0} consists of U_{inv0}^1 shown in Figure 13 and U_{inv0}^2 shown in Figure 14), which is used to compute the multiplicative inverse $(Ma)^{-1}$ in $F((2^4)^2)$, is no longer suitable for constructing c_2 . Therefore, we need to design a new quantum circuit $U_{inv1} : |p\rangle|h\rangle|0\rangle \rightarrow |p\rangle|h + p^{-1}\rangle|0\rangle$ to construct c_2 , and the quantum circuit for c_2 is introduced subsequently.

By combining the quantum circuits of $U_{q^2 \times \lambda}$ (see Figure 8), Mul_1 (see Figure 10), and $F(2^4)_{inv0}$ (see in Section III-B), the quantum circuit $U_{inv1}^1 : |p\rangle|h\rangle|0\rangle \rightarrow |p\rangle|h + p^{-1}\rangle|0\rangle$ for computing the multiplicative inverse in $F((2^4)^2)$ can be constructed as shown in Figure 16, where $F(2^4)_{inv0}^\dagger$ is the inverse process of $F(2^4)_{inv0}$.

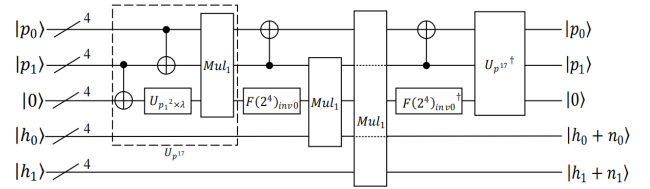
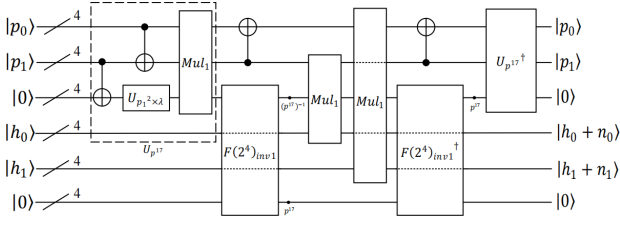


Fig. 16. The quantum circuit for $U_{inv1}^1 : |p\rangle|h\rangle|0\rangle \rightarrow |p\rangle|h + p^{-1}\rangle|0\rangle$.

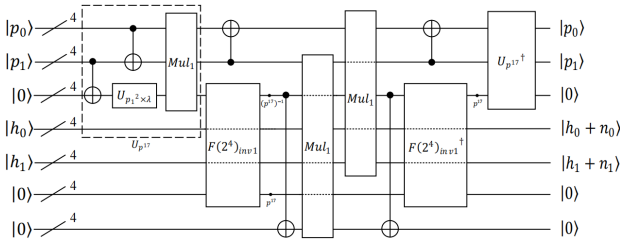
Similar to U_{inv0}^1 (as shown in Figure 13), if five clean ancillary qubits are added to the U_{inv1}^1 quantum circuit shown in Figure 16, the module $F(2^4)_{inv0}$ can be replaced with $F(2^4)_{inv1}$, resulting in U_{inv1}^2 (as shown in Figure 17). Although this replacement introduces 26 additional *Tofoli* gates, the *Tofoli* depth of the circuit is reduced by 8.

We observe that by adding 19 more clean ancillary qubits, the two Mul_1 in the U_{inv1}^2 shown in Figure 17 can be executed in parallel, resulting in U_{inv1}^3 (as shown in Figure

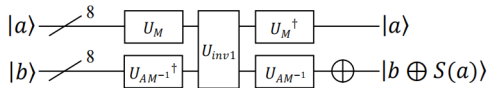
Fig. 17. The quantum circuit for $U_{inv1}^2 : |p\rangle|h\rangle|0\rangle \rightarrow |p\rangle|h + p^{-1}\rangle|0\rangle$.TABLE VIII
QUANTUM RESOURCE ESTIMATION FOR IMPLEMENTING c_2 .

Operation	#qubits	#Toffoli	#CNOT	#NOT	Toffoli depth
c_2^1	36	74	330	4	22
c_2^2	41	100	400	4	16
c_2^3	60	100	408	4	14

18). Although this replacement introduces 8 additional *CNOT* gates, the *Toffoli* depth of the circuit is reduced by 2.

Fig. 18. The quantum circuit for $U_{inv1}^3 : |p\rangle|h\rangle|0\rangle \rightarrow |p\rangle|h + p^{-1}\rangle|0\rangle$.

By combining the quantum circuits of U_M (see Figure 3), U_{inv1} (U_{inv1} consists of U_{inv1}^1 shown in Figure 16, U_{inv1}^2 shown in Figure 17, and U_{inv1}^3 shown in Figure 18), and $U_{AM^{-1}}$ (see Figure 4), the quantum circuit for $c_2 : |a\rangle|b\rangle|0\rangle \rightarrow |a\rangle|b \oplus S(a)\rangle|0\rangle$ can be constructed as shown in Figure 19, where $U_M^\dagger$ is the inverse process of U_M and $U_{AM^{-1}}^\dagger$ is the inverse process of $U_{AM^{-1}}$.

Fig. 19. The quantum circuit for $c_2 : |a\rangle|b\rangle|0\rangle \rightarrow |a\rangle|b \oplus S(a)\rangle|0\rangle$.

The quantum resources required for implementing c_2 are shown in Table VIII, where c_2^1 indicates that the U_{inv1} used in c_2 is U_{inv1}^1 , c_2^2 indicates that the U_{inv1} used in c_2 is U_{inv1}^2 , and c_2^3 indicates that the U_{inv1} used in c_2 is U_{inv1}^3 . The quantum resource estimation after decomposing the *Toffoli* gates into combinations of *T* gates and *Clifford* gates is summarized in Table IX.

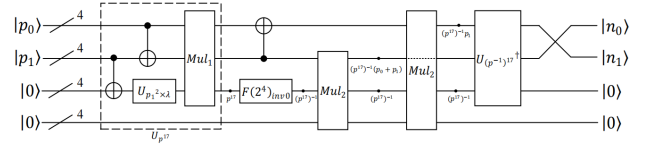
3) *The quantum circuit for $c_3 : |a\rangle|0\rangle \rightarrow |S(a)\rangle|0\rangle$* : In order to reduce the number of qubits used in the quantum circuit of the S-box, this section proposes a quantum circuit implementation for $c_3 : |a\rangle|0\rangle \rightarrow |S(a)\rangle|0\rangle$. Now, the quantum circuits for U_{inv0} (U_{inv0} consists of U_{inv0}^1 shown in Figure 13 and U_{inv0}^2 shown in Figure 14) and U_{inv1} (U_{inv1}

TABLE IX
QUANTUM RESOURCE ESTIMATION FOR IMPLEMENTING c_2 AFTER DECOMPOSING THE *Toffoli* GATES INTO COMBINATIONS OF *T* GATES AND *Clifford* GATES.

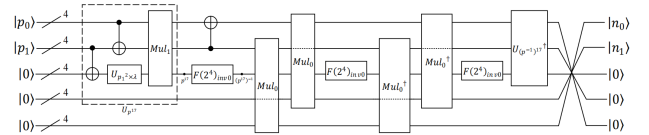
Operation	#qubits	#T	#Clifford	T depth
c_2^1	45	192	1142	16
c_2^2	49	224	1328	8
c_2^3	78	224	1336	7

consists of U_{inv1}^1 shown in Figure 16, U_{inv1}^2 shown in Figure 17, and U_{inv1}^3 shown in Figure 18), are no longer suitable for constructing c_3 . Therefore, we need to design a new quantum circuit $U_{inv2} : |p\rangle|0\rangle \rightarrow |p^{-1}\rangle|0\rangle$ to construct c_3 , and the quantum circuit for c_3 is introduced subsequently.

By combining the quantum circuits of $U_{q^2 \times \lambda}$ (see Figure 8), Mul_1 (see Figure 10) and Mul_2 (see Figure 11), and $F(2^4)_{inv0}$ (see in Section III-B), the quantum circuit $U_{inv2}^1 : |p\rangle|0\rangle \rightarrow |p^{-1}\rangle|0\rangle$ for computing the multiplicative inverse in $F((2^4)^2)$ can be constructed as shown in Figure 20.

Fig. 20. The quantum circuit for $U_{inv2}^1 : |p\rangle|0\rangle \rightarrow |p^{-1}\rangle|0\rangle$, where $p^{-1} = n_0 + n_1x$.

By examining the quantum circuits of Mul_2 (see Figure 11) and U_{inv2}^1 (see Figure 20), we observe that by adding four clean ancillary qubits, the two Mul_2 in U_{inv2}^1 can be reorganized, resulting in U_{inv2}^2 shown in Figure 21. This reorganization reduces the number of *Toffoli* gates and *Toffoli* depth by 14.

Fig. 21. The quantum circuit for $U_{inv2}^2 : |p\rangle|0\rangle \rightarrow |p^{-1}\rangle|0\rangle$.

Similar to U_{inv0} (U_{inv0} consists of U_{inv0}^1 shown in Figure 13 and U_{inv0}^2 shown in Figure 14) and U_{inv1} (U_{inv1} consists of U_{inv1}^1 shown in Figure 16, U_{inv1}^2 shown in Figure 17, and U_{inv1}^3 shown in Figure 18), if four clean ancillary qubits are added to the U_{inv2} (shown in Figure 21), the two $F(2^4)_{inv0}$ in U_{inv2}^2 can be replaced with $F(2^4)_{inv1}$ (shown in Figure 7), resulting in U_{inv2}^3 as shown in Figure 22. Although this replacement introduces 19 additional *Toffoli* gates, the *Toffoli* depth of the circuit is reduced by 15.

We observe that by adding 20 more clean ancillary qubits, the two Mul_1 in the U_{inv1} (shown in Figure 22) can be executed in parallel, resulting in U_{inv2}^4 shown in Figure 23. Although this replacement introduces 16 additional *CNOT* gates, the *Toffoli* depth of the circuit is reduced by 4.

By combining the quantum circuits of U_M (see Figure 3), U_{inv2} (U_{inv2} consists of U_{inv2}^1 shown in Figure 20, U_{inv2}^2

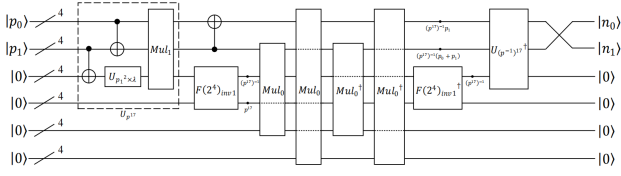
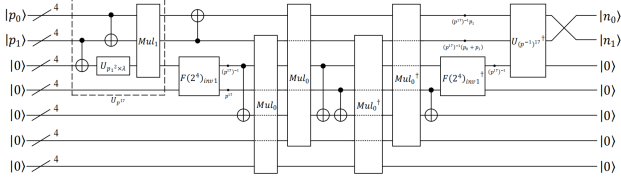
Fig. 22. The quantum circuit for $U_{inv2}^3 : |p\rangle|0\rangle \rightarrow |p^{-1}\rangle|0\rangle$.Fig. 23. The quantum circuit for $U_{inv2}^4 : |p\rangle|0\rangle \rightarrow |p^{-1}\rangle|0\rangle$.

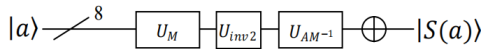
TABLE X
QUANTUM RESOURCE ESTIMATION FOR IMPLEMENTING c_3 .

Operation	#qubits	#Toffoli	#CNOT	#NOT	Toffoli depth
c_3^1	32	125	424	4	47
c_3^2	36	111	410	4	33
c_3^3	40	130	473	4	20
c_3^4	60	130	489	4	16

TABLE XI
QUANTUM RESOURCE ESTIMATION FOR IMPLEMENTING c_3 AFTER
DECOMPOSING THE Toffoli GATES INTO COMBINATIONS OF T GATES
AND Clifford GATES.

Operation	#qubits	#T	#Clifford	T depth
c_3^1	40	312	1860	36
c_3^2	44	264	1602	24
c_3^3	48	272	1659	10
c_3^4	76	272	1675	8

shown in Figure 21, U_{inv2}^3 shown in Figure 22, and U_{inv2}^4 shown in Figure 23) and U_{AM-1} (see Figure 4), the quantum circuit for $c_3 : |a\rangle|0\rangle \rightarrow |S(a)\rangle|0\rangle$ can be constructed as shown in Figure 24.

Fig. 24. The quantum circuit for $c_3 : |a\rangle|0\rangle \rightarrow |S(a)\rangle|0\rangle$.

The quantum resources required to implement c_3 are listed in Table X, where c_3^1 indicates that the U_{inv2} used in c_3 is U_{inv2}^1 , c_3^2 indicates that the U_{inv2} used in c_3 is U_{inv2}^2 , c_3^3 indicates that the U_{inv2} used in c_3 is U_{inv2}^3 , and c_3^4 indicates that the U_{inv2} used in c_3 is U_{inv2}^4 . The quantum resource estimation after decomposing the Toffoli gates into combinations of T gates and Clifford gates is summarized in Table XI.

IV. THE QUANTUM CIRCUIT FOR AES

In this section, we will utilize the designed quantum circuits for the S-box, along with the linear structures and key expansion process, to design the quantum circuit for AES.

A. The linear transformation structures of AES

The linear transformation structures of AES, ShiftRows, MixColumns and AddRoundKey, can all be implemented with CNOT gates only. The quantum resources required for these three linear transformations are detailed below.

1. AddRoundKey: Each AddRoundKey operation involves XORing a 128-bit subkey with the current state. This can be implemented in parallel using 128 CNOT gates.

2. ShiftRows: The ShiftRows operation only rearranges the output order of 16 bytes in the current state. It does not require any quantum operation for implementation.

3. MixColumns: The MixColumns operation processes 32 bits in the current state at a time and can be implemented using a 32×32 matrix. Xiang et al. [44] provides a quantum circuit for MixColumns operation with 92 CNOT gates. As MixColumns can be applied to all 4 columns of the state simultaneously, and the total requirement for CNOT gates is $92 \times 4 = 368$.

B. The key expansion process of AES

For the key expansion process, we implement the key expansion process of AES with Jaques et al.s method [26], but we use our S-box circuit for c_2 (see Figure 19). The quantum circuit as shown in Figure 25 represents the full structure of the key expansion process, denoted as $\mathcal{KE} : |k\rangle_{i-1}|0\rangle \rightarrow |k\rangle_i|0\rangle$. The SubBytes with an arrow in Figure 25 indicates that the state at the tail of the arrow first undergoes the SubBytes operation, and then the resulting value is then XORing with the state at the head of the arrow. The final result is assigned to the head of the arrow.

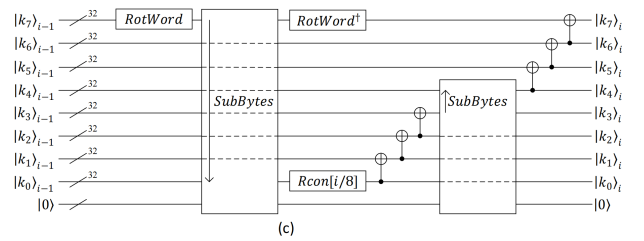
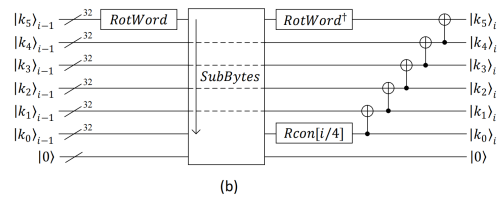
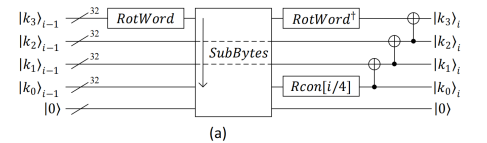


Fig. 25. Three fully structures of the key expansion process [26]. (a) A full structure of the key expansion process that can be used for AES-128. (b) A full structure of the key expansion process that can be used for AES-192. (c) A full structure of the key expansion process that can be used for AES-256.

The initial key of AES undergoes N_R rounds of the key expansion process, resulting in $(N_R + 1)$ round keys. Based

TABLE XII
QUANTUM RESOURCES REQUIRED TO IMPLEMENT THE KEY EXPANSION
PROCESSES OF AES.

	Operation	#qubits	#T	#Clifford	T depth
AES-128	c_2^1	244	7680	46659	160
	c_2^2	260	8960	54099	80
	c_2^3	376	8960	54419	70
AES-192	c_2^1	308	8448	51395	176
	c_2^2	324	9856	59579	88
	c_2^3	440	9856	59931	77
AES-256	c_2^1	372	9984	60639	208
	c_2^2	388	11648	70311	104
	c_2^3	504	11648	70727	91

on Algorithms 1 and 2, each round of key expansion process contains *RotWord*, *SubBytes*, *Rcon* and *XOR* four operations. The *RotWord* can be implemented by simply rearranging the output order of the qubits, without requiring any quantum operation. The *Rcon* is a constant (see Section II-A2) and can be efficiently implemented with at most four *NOT* gates for AES-128, at most five *NOT* gates for AES-192, while at most one *NOT* gate for AES-256 in each round of the key expansion process. The *XOR* operation between two 32-bit words during key expansion can be implemented with 32 *CNOT* gates. Since each round of the key expansion process operates on 128 qubits, only the relevant subset of the full structure of the key expansion process (see Figure 25) is needed, with other operations being neglected. Let $\mathcal{KE}_j^l$ represents the operation from $|k_j\rangle_{i-1}$ to $|k_l\rangle_{i-1}$ in $\mathcal{KE} : |k\rangle_{i-1}|0\rangle \rightarrow |k\rangle_i|0\rangle$, while ignoring other parts, where $0 \leq j \leq l$. The quantum resources required to implement the key expansion process of AES with *T* gates and *Clifford* gates are shown in Table XII.

C. The SubBytes operation of AES

The number of encryption rounds N_R for AES-128, AES-192, and AES-256 is 10, 12, and 14, respectively. Since the plaintext length is 128 bits, SubBytes operation of each round function requires 16 executions of the 8×8 S-box. Since this paper focuses on implementing the AES with as few qubits as possible, the SubBytes operation during encryption is implemented using $c_3 : |a\rangle|0\rangle \rightarrow |S(a)\rangle|0\rangle$. The quantum resources required to implement the quantum circuit for SubBytes in AES with *T* gates and *Clifford* gates are summarized in Table XIII.

D. Integrated quantum circuit for AES

The AES consists of the encryption process and the key expansion process. The encryption process includes an initial AddRoundKey operation followed by N_R rounds of round functions, and each round function consists of SubBytes, ShiftRows, MixColumns, and AddRoundKey. The quantum circuit of the AES is presented in Figure 26, where *SubBytes*, *ShiftRows* and *MixColumn* are the operations of SubBytes, ShiftRows and MixColumns, and $\mathcal{KE}_j^l$ represents the operation from $|k_j\rangle_{i-1}$ to $|k_l\rangle_{i-1}$ in $\mathcal{KE} : |k\rangle_{i-1}|0\rangle \rightarrow |k\rangle_i|0\rangle$, while disregarding other words.

TABLE XIII
QUANTUM RESOURCES REQUIRED TO IMPLEMENT THE SUBBYTES
OPERATION OF AES.

	Operation	#qubits	#T	#Clifford	T depth
AES-128	c_3^1	640	49920	297600	360
	c_3^2	704	42220	256320	240
	c_3^3	768	43520	265440	100
	c_3^4	1216	43520	268000	80
AES-192	c_3^1	704	59904	357120	432
	c_3^2	768	50688	307584	288
	c_3^3	832	52224	318528	120
	c_3^4	1280	52224	321600	96
AES-256	c_3^1	768	69888	416640	504
	c_3^2	832	59136	358848	336
	c_3^3	869	60928	371616	140
	c_3^4	1344	60928	375200	112

TABLE XIV
THE QUANTUM RESOURCES REQUIRED TO IMPLEMENT THE QUANTUM
CIRCUITS FOR AES-128, WHERE $DW(T)$ IS DEFINED AS THE PRODUCT OF
THE *T* DEPTH AND THE CIRCUIT WIDTH (I.E., THE NUMBER OF QUBITS),
AND $c_2^i - c_3^j$ INDICATES THAT c_2^i IS USED IN THE KEY EXPANSION AND c_3^j IS
USED IN THE SUBBYTES ($1 \leq i \leq 3$ AND $1 \leq j \leq 4$).

Operation	#qubits	#T	#Clifford	T depth	DW(T)
$c_2^1 - c_3^1$	884	57600	349219	360	318240
$c_2^2 - c_3^1$	900	58880	356659	360	324000
$c_2^3 - c_3^1$	1016	58880	356979	360	365760
$c_2^1 - c_3^2$	948	49920	307939	240	227520
$c_2^2 - c_3^2$	964	51200	315379	240	231360
$c_2^3 - c_3^2$	1080	51200	315699	240	259200
$c_2^1 - c_3^3$	1012	51200	317059	160	161920
$c_2^2 - c_3^3$	1028	52480	324499	100	102800
$c_2^3 - c_3^3$	1144	52480	324819	100	114400
$c_2^1 - c_3^4$	1460	51200	319619	160	233600
$c_2^2 - c_3^4$	1476	52480	327059	80	118080
$c_2^3 - c_3^4$	1592	52480	327379	80	127360

The quantum resources required to implement the SubBytes operation in the encryption process are described in Section IV-C, while those for ShiftRows, MixColumns and AddRoundKey are provided in Section IV-A. The quantum resources required for the key expansion process are detailed in Section IV-B. To sum up, the quantum resources required to implement the quantum circuit for AES-128, AES-192, and AES-256 with *T* gates and *Clifford* gates are summarized in Table XIV, Table XV, and Table XVI, respectively.

As shown in Tables XIV, XV, and XVI, the proposed quantum circuits of the S-box significantly reduce the quantum resources needed to implement the quantum circuit for AES. This optimization not only enhances the execution efficiency of the quantum circuit but also provides a feasible solution for implementing larger scale quantum encryption algorithms under limited quantum resources. To compare with other literatures, we select the case with the minimum *T* depth, and minimum $DW(T)$ value. The comparison results are shown in Table XVII.

As shown in Table XVII, the proposed quantum circuit for AES-128 in this work significantly reduce the $DW(T)$ value, while maintaining low number of qubits. According to the NIST guidelines for quantum circuit evaluation in Ref.[29],

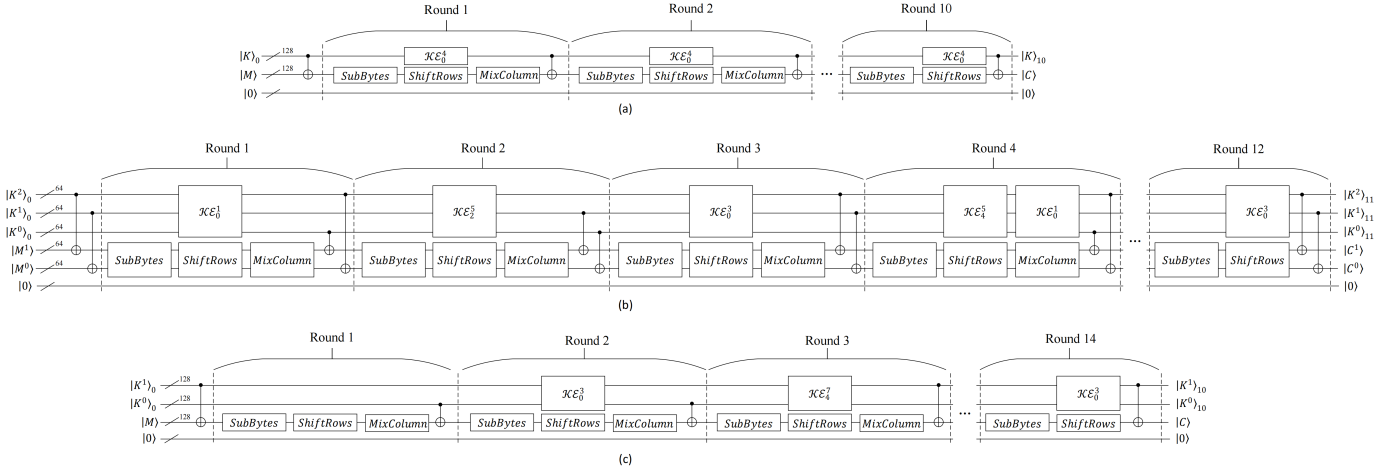


Fig. 26. The encryption process of AES. (a) The encryption process of AES-128. (b) The encryption process of AES-192. (c) The encryption process of AES-256.

TABLE XV

THE QUANTUM RESOURCES REQUIRED TO IMPLEMENT THE QUANTUM CIRCUITS FOR AES-192.

Operation	#qubits	# T	# $Clifford$	T depth	DW(T)
$c_2^1 \cdot c_3^1$	1012	68352	414467	432	437184
$c_2^2 \cdot c_3^1$	1028	69760	422651	432	444096
$c_2^3 \cdot c_3^1$	1144	69760	423003	432	494208
$c_2^1 \cdot c_3^2$	1076	59136	364931	288	309888
$c_2^2 \cdot c_3^2$	1092	60544	373115	288	314496
$c_2^3 \cdot c_3^2$	1208	60544	373467	288	347904
$c_2^1 \cdot c_3^3$	1140	60672	375875	176	200640
$c_2^2 \cdot c_3^3$	1156	62080	384059	120	138720
$c_2^3 \cdot c_3^3$	1272	62080	384411	120	152640
$c_2^1 \cdot c_3^4$	1588	60672	378947	176	279488
$c_2^2 \cdot c_3^4$	1604	62080	387131	96	153984
$c_2^3 \cdot c_3^4$	1720	62080	387483	96	165120

TABLE XVI

THE QUANTUM RESOURCES REQUIRED TO IMPLEMENT THE QUANTUM CIRCUITS FOR AES-256.

Operation	#qubits	# T	# $Clifford$	T depth	DW(T)
$c_2^1 \cdot c_3^1$	1140	79872	484223	504	574560
$c_2^2 \cdot c_3^1$	1156	81536	493895	504	582624
$c_2^3 \cdot c_3^1$	1272	81536	494311	504	641088
$c_2^1 \cdot c_3^2$	1204	69120	426431	336	404544
$c_2^2 \cdot c_3^2$	1220	70784	436103	336	409920
$c_2^3 \cdot c_3^2$	1336	70784	436519	336	448896
$c_2^1 \cdot c_3^3$	1268	70912	439199	208	263744
$c_2^2 \cdot c_3^3$	1284	72576	448871	140	179760
$c_2^3 \cdot c_3^3$	1400	72576	449287	140	196000
$c_2^1 \cdot c_3^4$	1716	70912	442783	208	356928
$c_2^2 \cdot c_3^4$	1732	72576	452455	112	193984
$c_2^3 \cdot c_3^4$	1848	72576	452871	112	206976

the depth and the DW(T) are considered key metrics for assessing the feasibility of quantum implementations. The depth and width of the quantum circuit are crucial for the performance of practical quantum devices. Compared with Ref.[42], our method achieves a lower DW(T) value with fewer qubits under the same T depth, reducing it by 20.09%.

TABLE XVII

COMPARISON OF THE QUANTUM CIRCUITS FOR AES-128.

Schemes	#qubits	T depth	DW(T)
Ours	1028	100	102800
[42]	1608	80	128640
[41]	3689	40	147560
[39]	5576	60	334560
[37]	256	29490	7549440

Moreover, our method also significantly outperforms Ref.[41] and Ref.[39] in terms of the DW(T) value. Compared with Ref.[37], although our approach uses more qubits, the DW(T) value is reduced by 98.64%. Unlike Refs.[37], [39], [41], which reduce circuit width at the cost of increased T depth (or vice versa), our AES quantum circuits achieve the lowest DW(T) value, demonstrating a more efficient depth-width trade-off.

V. CONCLUSION

This paper presents AES quantum circuits realized with lower DW(T). We improved upon the method proposed in Ref.[35] for quantum circuit design, optimizing the AES S-box quantum circuit design by introducing the composite field $F((2^4)^2)$, which is decomposed into a sequence of operations over the smaller finite field $F(2^4)$.

Specifically, we optimize matrix multiplication by reducing the number of $CNOT$ gates, lowering the number of T gates and T depth in the inversion circuit in $F(2^4)$, and achieving the quantum multiplication circuit in $F(2^4)$ with a reduced width and depth. Combined with the key expansion process, we achieved the lowest known DW(T) value to date, which is 102800.

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